



DEPARTMENT OF THE ARMY
U.S. ARMY CORPS OF ENGINEERS
HUMPHREYS ENGINEER CENTER SUPPORT ACTIVITY
7701 TELEGRAPH ROAD
ALEXANDRIA, VA 22315-3860

REPLY TO
ATTENTION OF

Contracting Office: CEHEC-CT

08 MAY 2025

**SOLE SOURCE MEMORANDUM FOR ACTIONS UNDER THE
SIMPLIFIED ACQUISITION THRESHOLD IAW FAR 13.106-1(b)(1)(i)**

TITLE: HyperMesh Software for US Army Corps of Engineers (USACE) Institute for Water Resources (IWR) Risk Management Center (RMC)

1. Contracting Activity: The USACE, Humphreys Engineer Center Support Activity (HECSA)

2. Description of Action: Requesting approval for a new Firm Fixed Price contract using Civil Workfunding (96 x 4902), The United States Army Corps of Engineers (USACE).

3. Description of Supplies/Services: IWR would like to purchase software to analyze potential problems and determine risk and plot solutions on Dam and Levee projects. CEIWR-RMC is requesting to purchase sixty-three (63) units of a 1-year subscription to Altair Engineering, Inc's (Altair) HyperMesh software. Altair® HyperMesh® is a powerful simulation solution that redefines CAE design processes across industries with unmatched pre- and post-processing capabilities for finite element modeling. Powered by AI-augmented 3D modeling and visualization tools, next-gen design and optimization workflows, and an open, programmable, user-friendly interface, HyperMesh empowers users to effortlessly manage large and complex models, unveil critical insights, optimize designs, and foster innovation. It revolutionizes engineering workflows with unparalleled precision to enhance productivity and scale operations.

Altair is an American multinational information technology company headquartered in Troy, Michigan. It provides software and cloud solutions for simulation, IoT, high performance computing (HPC), data analytics, and artificial intelligence (AI). Altair is the creator of the Hyper Mesh software product, among numerous other software packages and suites.

HyperMesh will provide simulation and design technology that encompasses all disciplines – from structures to motion, fluids, thermal management, electromagnetics, electronic system design, system modeling, and more – to give users the power to build better products and processes faster. HyperMesh will reduce development times, lower costs, reduce waste and energy usage, and streamline the entire product lifecycle from start to finish, so users know they're optimizing everything from concept to in-service operation and beyond.

Altair's Enterprise Units model provides dynamic and fluid access to the software pool. While the initial allocation assumes sixty-three (63) units per user, the flexible licensing structure allows for the optimization of concurrent user demand:

- Users share access to a centralized pool of 63 units, ensuring that usage scales effectively across teams.

- Depending on how concurrent user activity aligns, the per-user unit demand may decrease as multiple users leverage the larger shared pool at different times.
- This scalability reduces the need to over-purchase licenses, enabling cost savings while maintaining seamless access to tools like Hyper Mesh.
- Altair’s approach maximizes value by ensuring that resources dynamically adapt to varying project demands without restricting individual user access.

Period of Performance/Delivery Date: The contract will include a base year of twelve (12) months with two (2) option periods of twelve (12) months, totaling thirty-six (36) months, totaling approximate cost of \$xxx,xxx.xx.

Item#/MFG. Part#	Quantity	Unit price	Total Cost
Altair Units - Enterprise Suite - Lease AU-ENT-STD-AN Base Year	63	\$xxx.xx	\$xx,xxx.xx
Option Year 1	63	\$xxx.xx	\$xx,xxx.xx
Option Year 2	63	\$x,xxx.xx	\$xx,xxx.xx

4. Authority Cited: 10 U.S.C. 2304(C)(1) or 41 U.S.C. 253(C)(1) as implemented in FAR 13.106-1(b)(1)(i)

5. Reason for Authority Cited: The Government must purchase this specific HyperMesh software and its components because RMC staff has already been trained on this specific software and has already used a trial license to ensure it would be successful in running the modeling applications required by structural engineers. Also, this software has already been approved for use on USACE computers and is compatible with the CorpsNet network. If there is a purchase of software other than this specific software, then all RMC personnel would need to be retrained on the new software, and the new software would need to go through ACE-IT's lengthy approval process to be installed. This delay would cause project milestones to be missed, and the retraining of staff would cause a significant increase in cost to the Government. This could range between \$100k-\$150k which stems from a collective need to have 800 hours of labor for all RMC personnel to be retrained on new software. Further, any different software would require approval for use on USACE computers and the CorpsNet network which would further add to delays. The software manger also searched for other comparable software while conducting market research and there was not any other software that had any of the same capabilities as Altair’s HyperMesh.

6. Efforts to Obtain Competition: The requesting organization has provided a statement of non-availability from the Computer Hardware, Enterprise Software and Solutions (CHESS) office and/or vendor quote for contracted services in accordance with the Secretary of the Army Waiver Process for Commercial-Off-the-Shelf Information Technology (COTS-IT) Procurement Outside the CHESS Program.

On 29 APR 2025, Altair confirmed they do not use resellers for the company’s products.

On 03 SEP 2024, The Army Deputy Chief of Staff (DCS), G6 has validated the subject requirement, and this memorandum serves as official approval of the IT Approval System Waiver indicated. The waiver approval number is: 20240617093252247.

On 05 JUN 2024, the IWR RMC technical team (customer) posted a Request for Quote #494176 on the CHESS SW2 contract vehicle to see if this software was available on CHESS. The results of the customer posted RFQ are as follows:

- DH Technologies, Inc. / No bid at this time.
- Enterprise Technology Solutions, Inc / No bid at this time.
- Govplace, Inc. / No bid at this time.
- Carahsoft Technology Corporation / No bid at this time.
- Vertosoft LLC / No bid at this time.
- CDW Government LLC / No bid at this time.
- SHI International Corp / No bid at this time.
- DLT Solutions / No bid at this time.
- Epoch Concepts, LLC / No bid at this time.
- Thunder Cat Technology LLC / No bid at this time.
- Blazar Technology Solutions / No bid at this time.
- Blue Tech, Inc. / No bid at this time.
- Red River Technology LLC / No bid at this time.
- York Telecom Corporation / No bid at this time.
- Insight Public Sector, Inc. / No bid at this time.
- Four LLC, dba Four Inc / No bid at this time.
- Immix Technology, Inc / No bid at this time.
- Gov Connection, Inc. / No bid at this time.
- Dynamic Systems, Inc / No bid at this time.
- Bow Technologies, LLC / No bid at this time.
- World Wide Technology, LLC / No bid at this time.
- ID Technologies, LLC / No bid at this time.
- Strategic Communications, LLC / No bid at this time.
- Counter Trade Products, Inc / No bid at this time.

7. Actions to Increase Competition: USACE has and will continue to evaluate products from other vendors in the market to determine whether a comparable product is produced in the future, prior to any award. At the current time, there are no other products which will meet the requirements of USACE.

8. Market Research: As detailed above with respect to the customer's RFQ, there are no authorized resellers to provide this product. Altair, the software developer, is the only authorized provider of HyperMesh.

9. Interested Sources: To date, no other sources have written to express an interest. Notices required by FAR 5.201 shall be published and any bids and proposals received shall be considered.

10. Other Facts: It would be prohibitively expensive for another company to develop competitive forecasting services solely to meet IWR needs and specifications.

11. Fair and Reasonable Cost Determination: I hereby determine that the anticipated price to the Government for this contract action will be fair and reasonable, based on comparison to an Independent Government Cost Estimate. Per FAR 15.403-1(b)(3), certified cost and pricing data is not required.

David A. Kaplan
Procuring Contracting Officer
Phone: 703-428-8487

12. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

David A. Kaplan
Procuring Contracting Officer
Phone: 703-428-8487

Approval

Based on the foregoing justification, I hereby approve an award of a firm fixed price contract to Altair Engineering Inc. in the amount of \$xxx,xxx.xx on an other than full and open competition basis pursuant to the authority of FAR 13.106-1(b)(1)(i), subject to availability of funds, and provided that the services and property herein described have otherwise been authorized for acquisition.

David A. Kaplan
Procuring Contracting Officer
Phone: 703-428-8487

Prepared by:

Bart Dziadosz
Contract Specialist